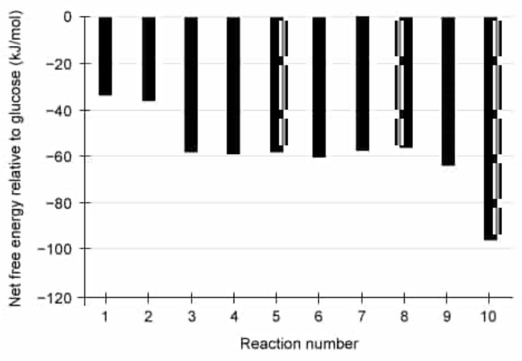
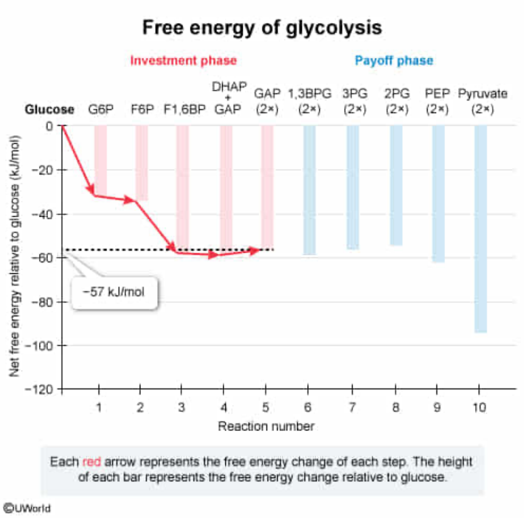


The graph above shows the free energy of the products of each step of glycolysis relative to the free energy of glucose under typical cellular conditions (free energy of glucose is set to zero). Based on the data shown, the total free energy change during the energy investment reactions of glycolysis, in which two molecules of glyceraldehyde-3-phosphate are formed, is closest to which of the following?

- A. −245 kJ/mol [14%]
- B. −95 kJ/mol [12%]
- C. −57 kJ/mol [67%]
- D. −39 kJ/mol [6%]

Correct
68% answered correctly

Explanation:



Glycolysis is the series of ten cytosolic reactions by which a glucose molecule is metabolized into two pyruvate molecules, generating ATP and the reduced electron carrier NADH in the process. **Glycolysis consists of** successive **energy investment** (ie, the **first five reactions**) and **energy payoff** (ie, the **last five reactions**) phases during which two ATP molecules are invested and four ATP molecules are synthesized, respectively.

Each reaction of glycolysis is accompanied by a free energy change, which determines whether the reaction is spontaneous or not. Reactions with small free energy changes are considered reversible whereas those with larger free energy changes are considered irreversible (ie, a considerable amount of energy would need to be added to drive the reaction in the opposite direction).

The graph in the question depicts the free energy changes that occur during the reactions of glycolysis, depicted as the net free energy change relative to the original free energy of glucose. The question asks for the cumulative free energy change during the energy investment reactions of glycolysis, in which glyceraldehyde-3-phosphate is formed (ie, the first five reactions). Correct reading of the **graph indicates** that the **cumulative free energy** change **after** the **fifth reaction** of glycolysis **is about −57 kJ/mol**.

(Choice A) −245 kJ/mol is the sum of the cumulative free energy changes after each of the reactions of the energy investment phase of glycolysis. Summing would be the correct approach if the graph depicted the *individual* free energy changes associated with each reaction. However, the graph depicts the cumulative net change in free energy relative to glucose, so each bar already incorporates the individual free energy changes associated with all previous reactions.

(Choice B) −95 kJ/mol is the approximate free energy change after the *payoff* phase of glycolysis (ie, after the tenth reaction).

(Choice D) −39 kJ/mol is the approximate free energy change occurring during the *payoff* phase alone (ie, the free energy of the fifth reaction products subtracted from the free energy of the tenth reaction products).

Educational objective:
The reactions of glycolysis are accompanied by free energy changes that determine the spontaneity and reversibility of the reactions. The first five reactions of glycolysis are the energy investment phase, and the second five reactions are the energy payoff phase.

Time spent: 0 sec	**QID**:403210
Subject: Biochemistry	**System:** Metabolic Reactions